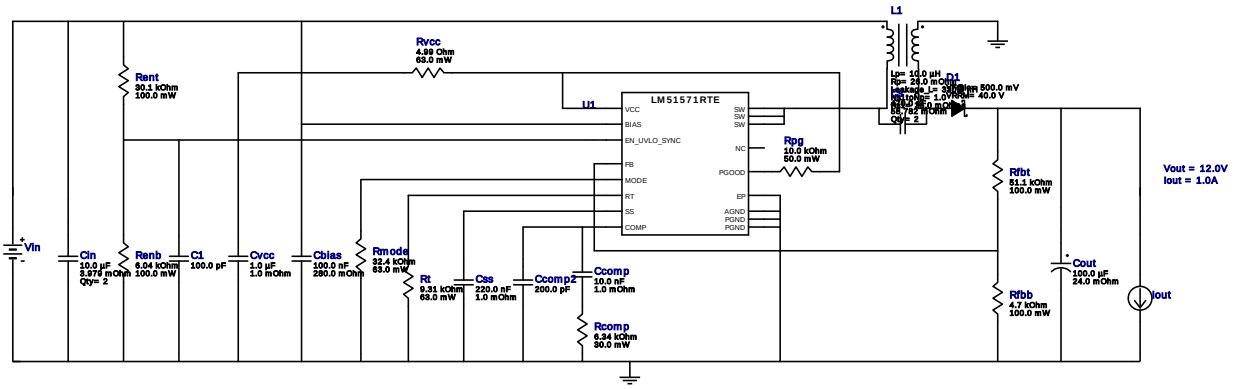
WEBENCH® Design Report

VinMin = 9.0V
VinMax = 16.0V
Vout = 12.0V
Iout = 1.0A

Device = LM5157RTER
Topology = SEPIC
Created = 2026-10-13 07:22:21.933
BOM Cost = NA
BOM Count = 23
Total Pd = 1.34W

Design : 8 LM5157RTER
LM5157RTER 9V-16V to 12.00V @ 1A

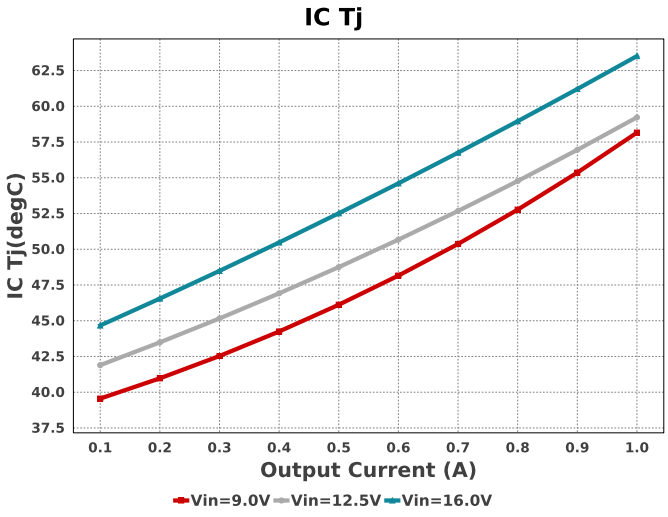


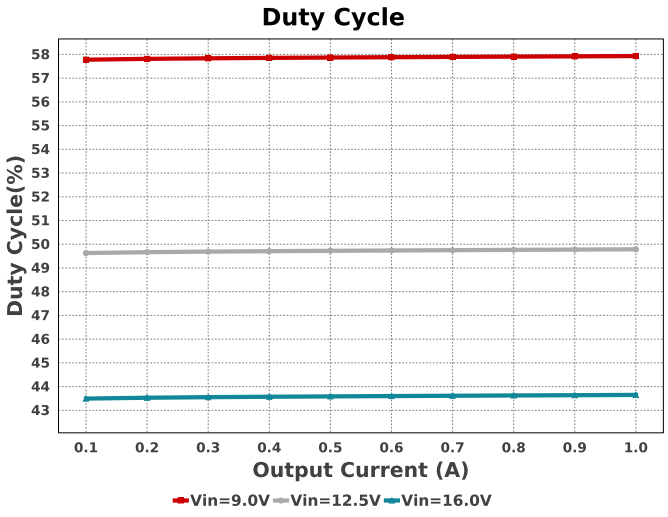
Electrical BOM

Name	Manufacturer	Part Number	Properties	Qty	Price	Footprint
C1	Samsung Electro-Mechanics	CL21C101JBANNNC Series= C0G/NP0	Cap= 100.0 pF VDC= 50.0 V IRMS= 0.0 A	1	$0.01	0805 7 mm²
Cbias	AVX	08053C104KAT2A Series= X7R	Cap= 100.0 nF ESR= 280.0 mOhm VDC= 25.0 V IRMS= 0.0 A	1	$0.01	0805 7 mm²
Ccomp	MuRata	GRM155R71C103KA01D Series= X7R	Cap= 10.0 nF ESR= 1.0 mOhm VDC= 16.0 V IRMS= 0.0 A	1	$0.01	0402 3 mm²
Ccomp2	Samsung Electro-Mechanics	CL10C201JB8NNNC Series= C0G/NP0	Cap= 200.0 pF VDC= 50.0 V IRMS= 0.0 A	1	$0.01	0603 5 mm²
Cin	Taiyo Yuden	MSASU32MSB5106KPNA01 Series= X5R	Cap= 10.0 uF ESR= 3.979 mOhm VDC= 50.0 V IRMS= 3.4821 A	2	$0.16	1210 15 mm²
Cout	Panasonic	25SVPF100M Series= SVPF	Cap= 100.0 uF ESR= 24.0 mOhm VDC= 25.0 V IRMS= 3.2 A	1	$0.66	CAPSMT_62_E7 106 mm²
Cs	Taiyo Yuden	MSAST219SB5474MTNA01 Series= X5R	Cap= 470.0 nF ESR= 58.782 mOhm VDC= 25.0 V IRMS= 925.89 mA	2	$0.02	0805 7 mm²
Css	Kemet	C0603C224Z4VACTU Series= Y5V	Cap= 220.0 nF ESR= 1.0 mOhm VDC= 16.0 V IRMS= 0.0 A	1	$0.01	0603 5 mm²
Cvcc	Taiyo Yuden	EMK107B7105KA-T Series= X7R	Cap= 1.0 uF ESR= 1.0 mOhm VDC= 16.0 V IRMS= 0.0 A	1	$0.01	0603 5 mm²
D1	Fairchild Semiconductor	SSA24	VF@Io= 500.0 mV VRRM= 40.0 V	1	$0.17	SMA 37 mm²

Copyright © 2026, Texas Instruments Incorporated

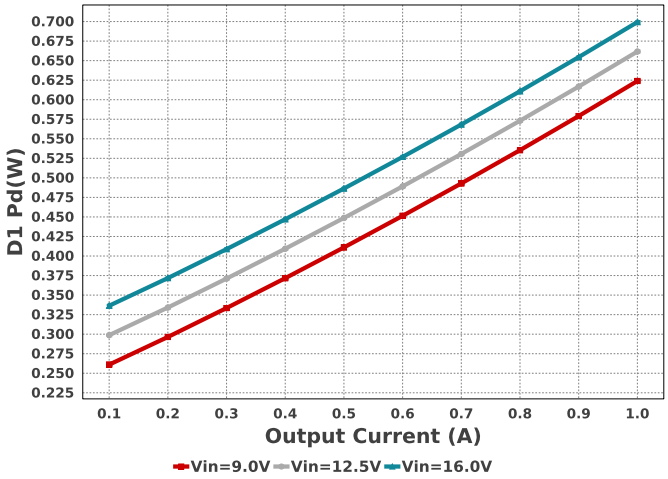
Name	Manufacturer	Part Number	Properties	Qty	Price	Footprint
L1	Coilcraft	MSD1583-103MEB	Lp= 10.0 µH Rp= 26.0 mOhm Leakage_L= 330.0 nH Ns1toNp= 1.0 Rs1= 26.0 mOhms	1	NA	IND_MSD1583 282 mm²
Rcomp	Delta Electronics Inc.	PFR02S-6341-FNH Series= ?	Res= 6.34 kOhm Power= 30.0 mW Tolerance= 1.0%	1	$0.01	01005 2 mm²
Renb	Susumu Co Ltd	RG1608P-6041-B-T5 Series= RG1608	Res= 6.04 kOhm Power= 100.0 mW Tolerance= 0.1%	1	$0.06	0603 5 mm²
Rent	Yageo	RT0603BRD0730K1L Series= ?	Res= 30.1 kOhm Power= 100.0 mW Tolerance= 0.1%	1	$0.04	0603 5 mm²
Rfbb	Susumu Co Ltd	RG1608P-472-B-T5 Series= RG1608	Res= 4.7 kOhm Power= 100.0 mW Tolerance= 0.1%	1	$0.04	0603 5 mm²
Rfbt	Susumu Co Ltd	RG1608P-5112-B-T5 Series= RG1608	Res= 51.1 kOhm Power= 100.0 mW Tolerance= 0.1%	1	$0.06	0603 5 mm²
Rmode	Vishay-Dale	CRCW040232K4FKED Series= CRCW..e3	Res= 32.4 kOhm Power= 63.0 mW Tolerance= 1.0%	1	$0.01	0402 3 mm²
Rpg	Yageo	RC0201FR-0710KL Series= ?	Res= 10.0 kOhm Power= 50.0 mW Tolerance= 1.0%	1	$0.01	0201 2 mm²
Rt	Vishay-Dale	CRCW04029K31FKED Series= CRCW..e3	Res= 9.31 kOhm Power= 63.0 mW Tolerance= 1.0%	1	$0.01	0402 3 mm²
Rvcc	Vishay-Dale	CRCW04024R99FKED Series= CRCW..e3	Res= 4.99 Ohm Power= 63.0 mW Tolerance= 1.0%	1	$0.01	0402 3 mm²
U1	Texas Instruments	LM5157RTER	Switcher	1	$1.50	RTE0016K 16 mm²



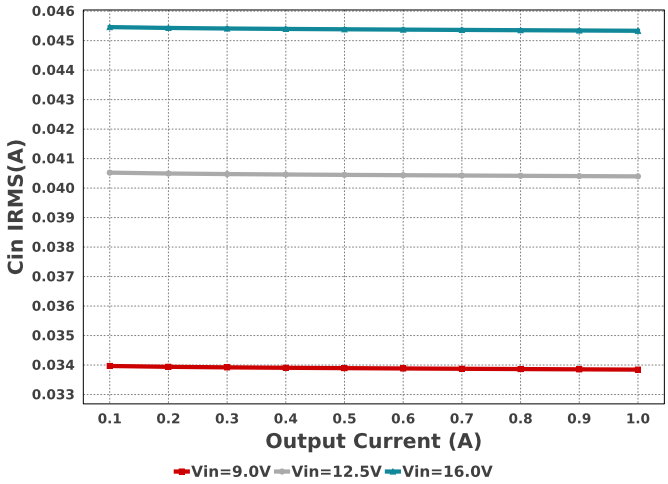


Copyright © 2026, Texas Instruments Incorporated

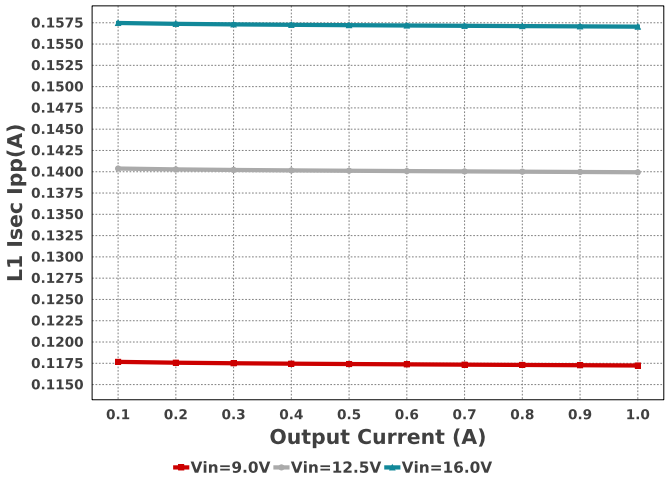
D1 Pd



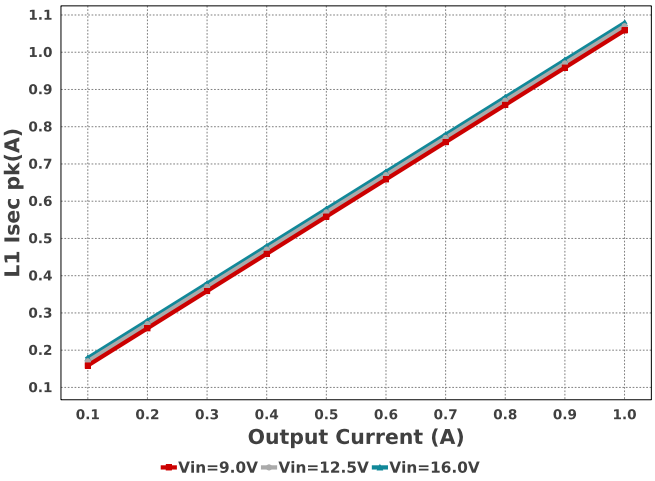
Cin IRMS



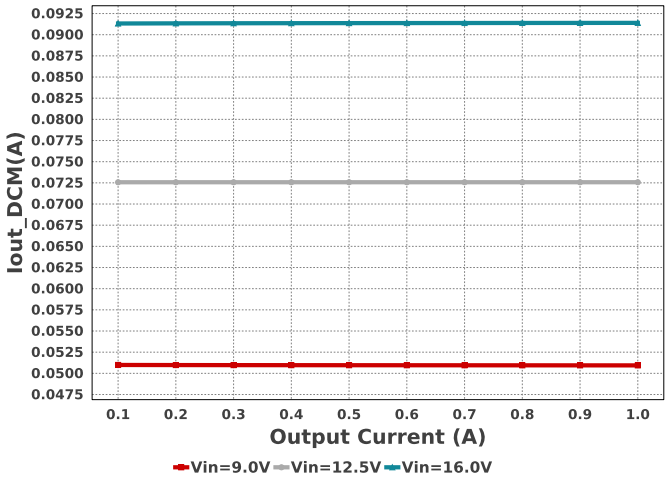
L1 Isec Ipp



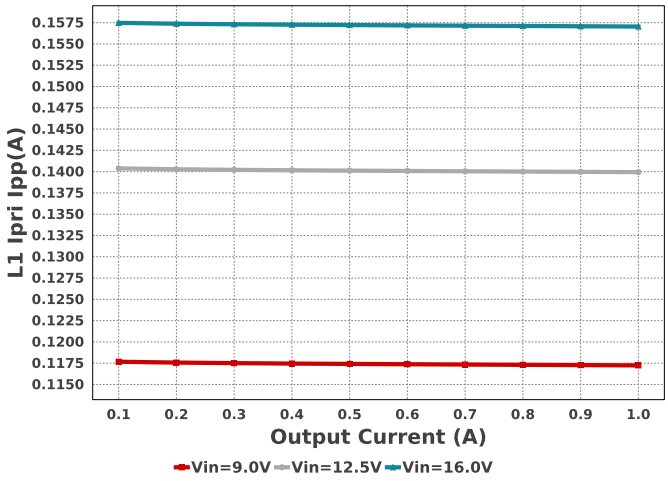
L1 Isec pk



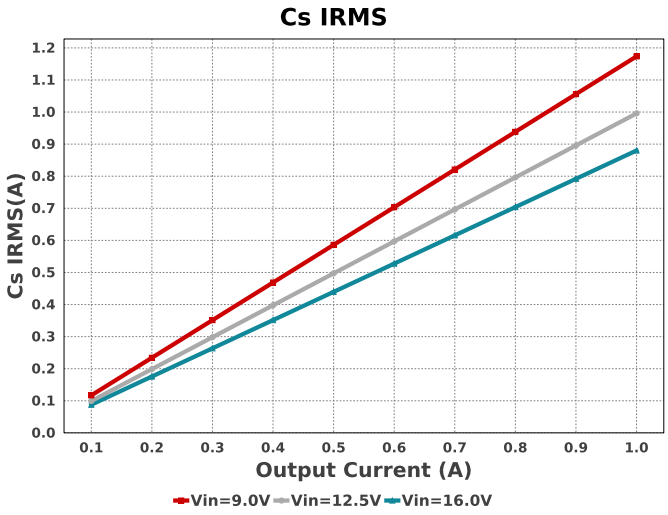
Iout_DCM

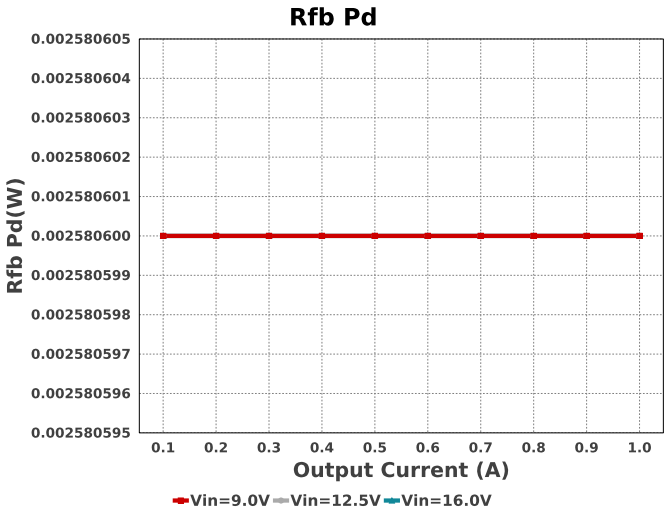


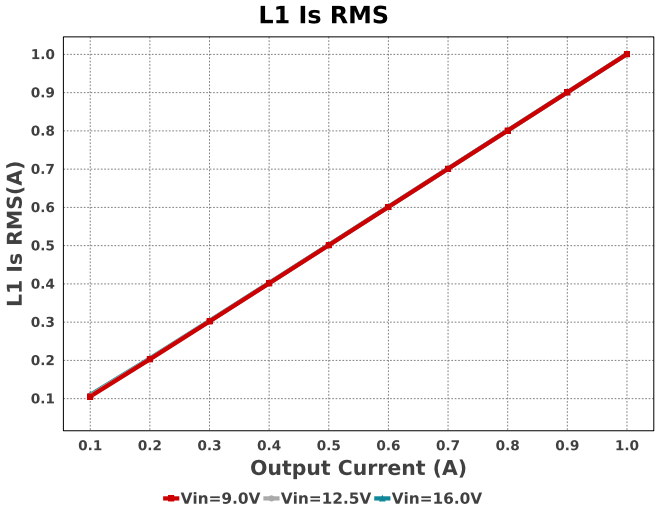
L1 Ipri Ipp

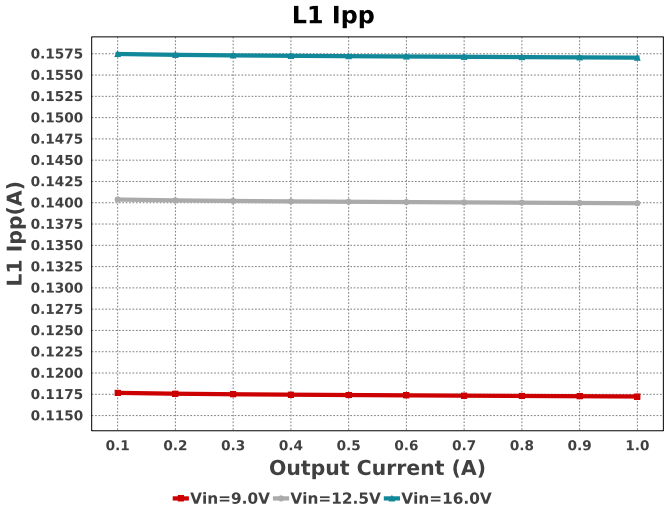


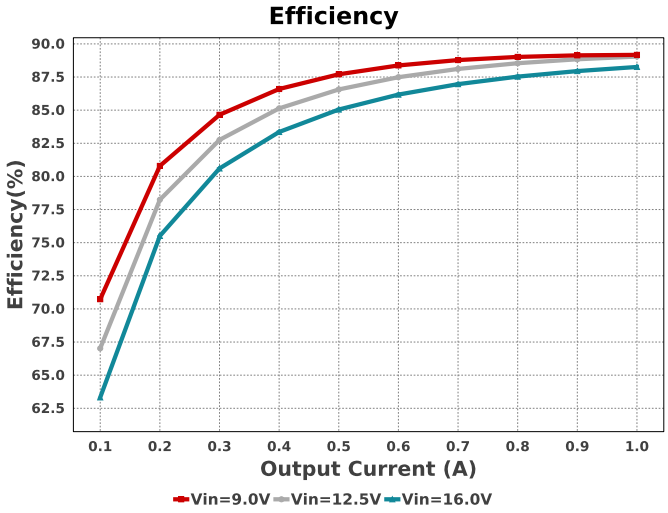
Copyright © 2026, Texas Instruments Incorporated

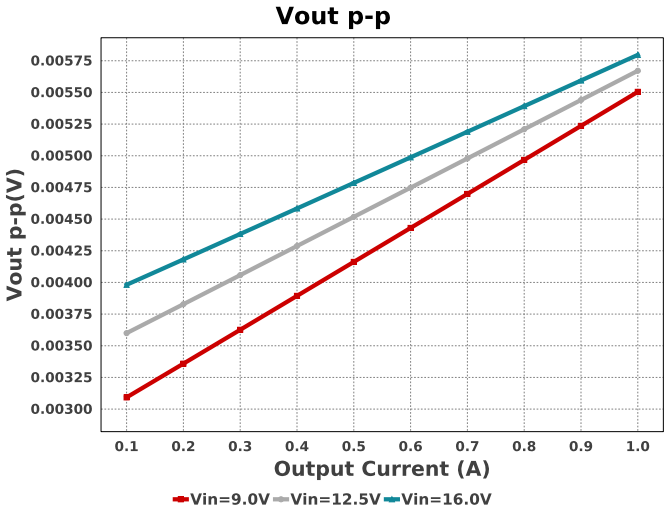




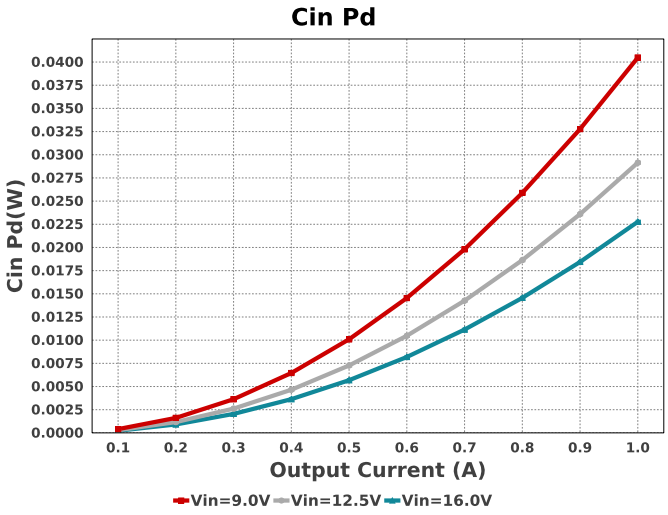


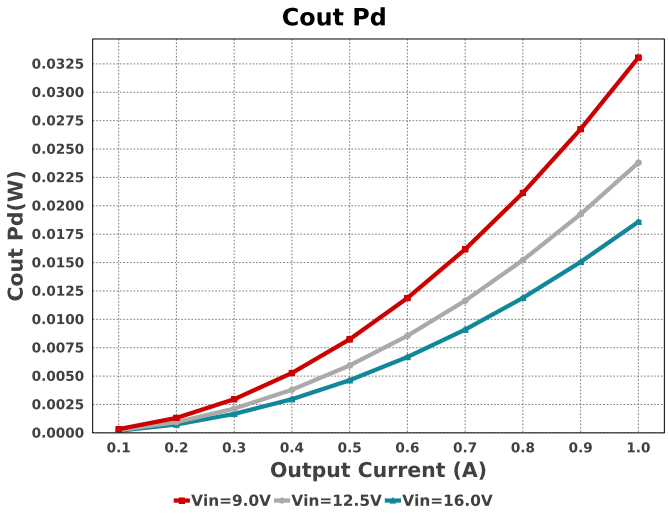


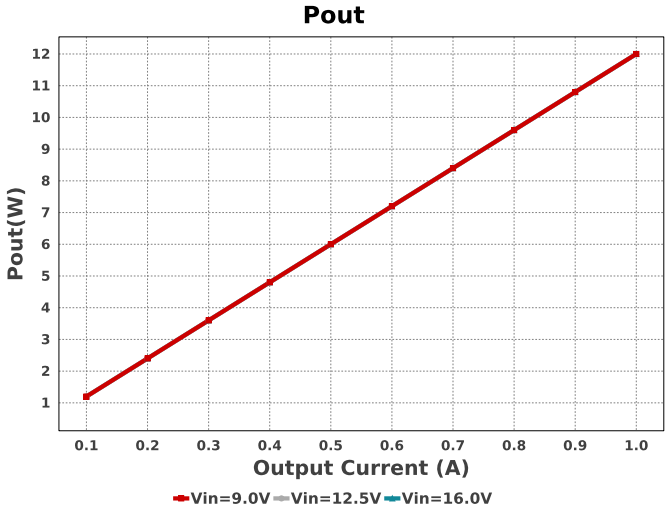


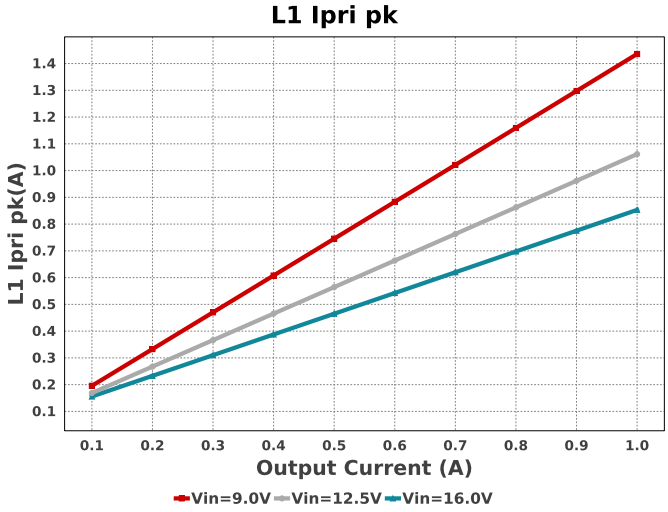


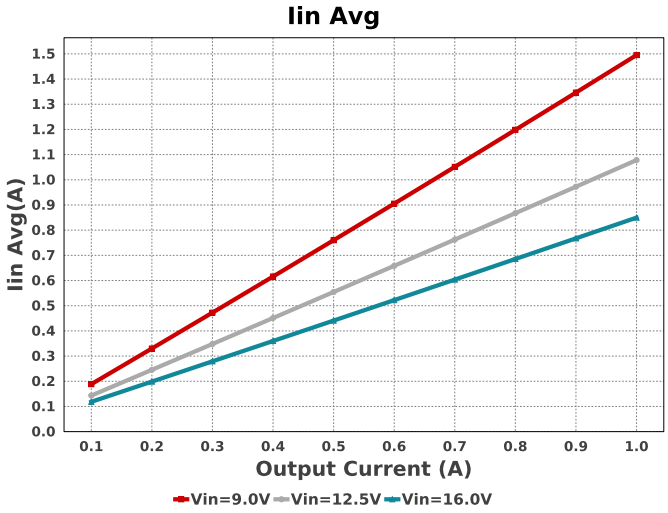
Copyright © 2026, Texas Instruments Incorporated

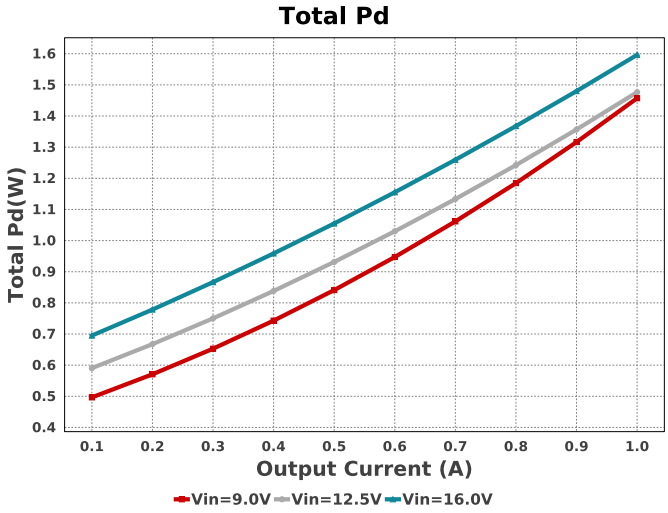




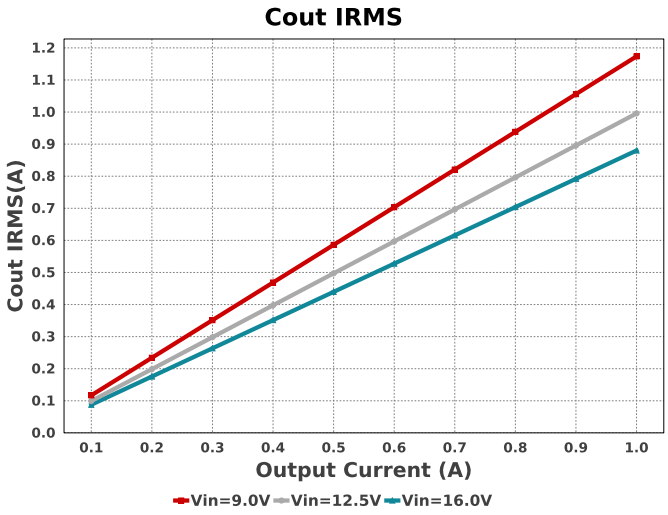


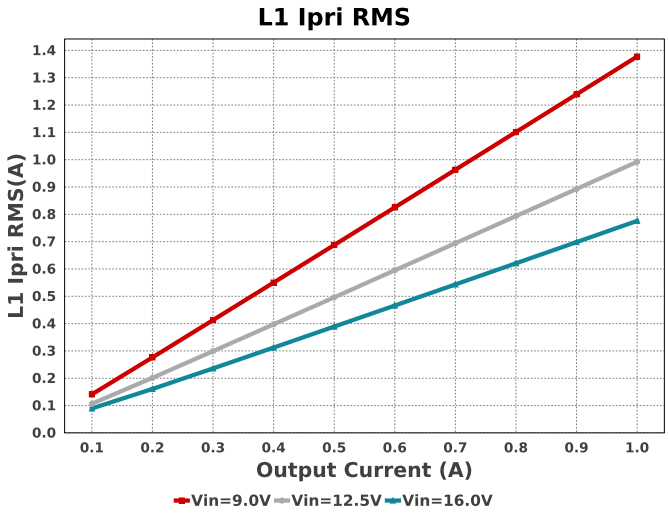


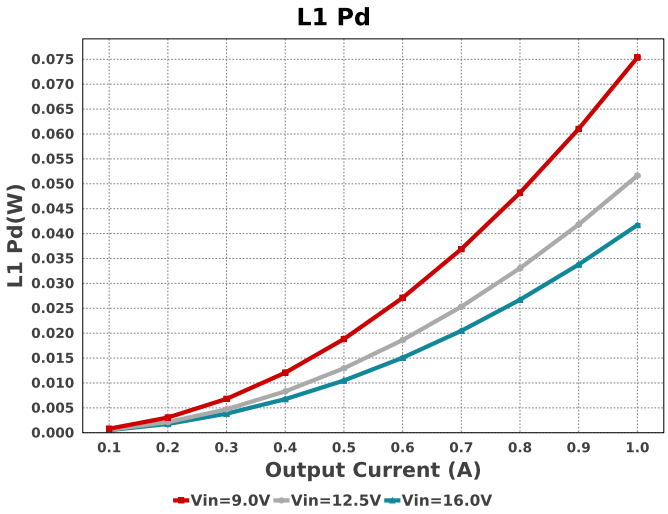


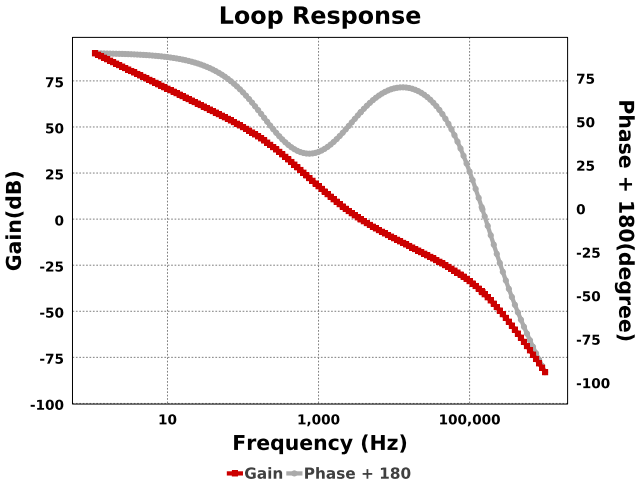


Copyright © 2026, Texas Instruments Incorporated









Operating Values

#	Name	Value	Category	Description
1.	Cin IRMS	33.844 mA	Capacitor	Input capacitor RMS ripple current
2.	Cin Pd	40.476 mW	Capacitor	Input capacitor power dissipation
3.	Cout IRMS	1.173 A	Capacitor	Output capacitor RMS ripple current
4.	Cout Pd	33.05 mW	Capacitor	Output capacitor power dissipation
5.	Cs IRMS	1.173 A	Capacitor	Coupling capacitor RMS ripple current
6.	D1 Pd	623.95 mW	Diode	Output Diode Power Dissipation
7.	IC Tj	53.119 degC	IC	IC junction temperature
8.	ICThetaJA	41.3 degC/W	IC	IC junction-to-ambient thermal resistance
9.	Iin Avg	1.482 A	IC	Average input current
10.	Cin Pd	40.476 mW	Power	Input capacitor power dissipation
11.	Cout Pd	33.05 mW	Power	Output capacitor power dissipation
12.	D1 Pd	623.95 mW	Power	Output Diode Power Dissipation
13.	L1 Pd	75.364 mW	Power	Power Dissipation in the Inductor
14.	Rfb Pd	2.581 mW	Power	Rfb Power Dissipation
15.	Total Pd	1.335 W	Power	Total Power Dissipation
16.	Rfb Pd	2.581 mW	Resistor	Rfb Power Dissipation
17.	BOM Count	23	System Information	Total Design BOM count
18.	Cross Freq	3.598 kHz	System Information	Bode plot crossover frequency
19.	Duty Cycle	57.931 %	System Information	Duty cycle
20.	Efficiency	89.987 %	System Information	Steady state efficiency
21.	FootPrint	547.0 mm^2	System Information	Total Foot Print Area of BOM components
22.	Frequency	2.153 MHz	System Information	Switching frequency
23.	Frequency	2.153 MHz	System Information	Switching frequency
24.	Gain Marg	-39.615 dB	System Information	Bode Plot Gain Margin

Copyright © 2026, Texas Instruments Incorporated

#	Name	Value	Category	Description
25.	Iout	1.0 A	System Information	Iout operating point
26.	Iout	1.0 A	System Information	Iout operating point
27.	Iout_DCM	50.939 mA	System Information	Approximate Current below which DCM mode of operation will begin
28.	Low Freq Gain	89.957 dB	System Information	Gain at 1Hz
29.	Mode	CCM	System Information	Conduction Mode
30.	Phase Marg	55.791 deg	System Information	Bode Plot Phase Margin
31.	Pout	12.0 W	System Information	Total output power
32.	Total BOM	NA	System Information	Total BOM Cost
33.	Vin	9.0 V	System Information	Vin operating point
34.	Vin	9.0 V	System Information	Vin operating point
35.	Vout	12.0 V	System Information	Operational Output Voltage
36.	Vout Actual	11.872 V	System Information	Vout Actual calculated based on selected voltage divider resistors
37.	Vout Tolerance	1.185 %	System Information	Vout Tolerance based on IC Tolerance (no load) and voltage divider resistors if applicable
38.	Vout p-p	5.505 mV	System Information	Peak-to-peak output ripple voltage
39.	L1 Ipp	117.24 mA	Transformer	Inductor 1 peak to peak current
40.	L1 Ipri Ipp	117.24 mA	Transformer	L1 primary side ripple pk-pk.
41.	L1 Ipri RMS	1.377 A	Transformer	L1 primary rms current.
42.	L1 Ipri pk	1.436 A	Transformer	L1 primary pk current.
43.	L1 Is RMS	1.001 A	Transformer	L1 secondary rms current
44.	L1 Isec Ipp	117.24 mA	Transformer	L1 secondary side ripple pk-pk.
45.	L1 Isec pk	1.059 A	Transformer	Peak current in L1 secondary
46.	L1 Pd	75.364 mW	Transformer	Power Dissipation in the Inductor

Design Inputs

Name	Value	Description
Iout	1.0	Maximum Output Current
VinMax	16.0	Maximum input voltage
VinMin	9.0	Minimum input voltage
Vout	12.0	Output Voltage
base_pn	LM51571	Base Product Number
source	DC	Input Source Type
Ta	30.0	Ambient temperature

Copyright © 2026, Texas Instruments Incorporated

WEBENCH® Assembly

Component Testing

Some published data on components in datasheets such as Capacitor ESR and Inductor DC resistance is based on conservative values that will guarantee that the components always exceed the specification. For design purposes it is usually better to work with typical values. Since this data is not always available it is a good practice to measure the Capacitance and ESR values of Cin and Cout, and the inductance and DC resistance of L1 before assembly of the board. Any large discrepancies in values should be electrically simulated in WEBENCH to check for instabilities and thermally simulated in WebTHERM to make sure critical temperatures are not exceeded.

Soldering Component to Board

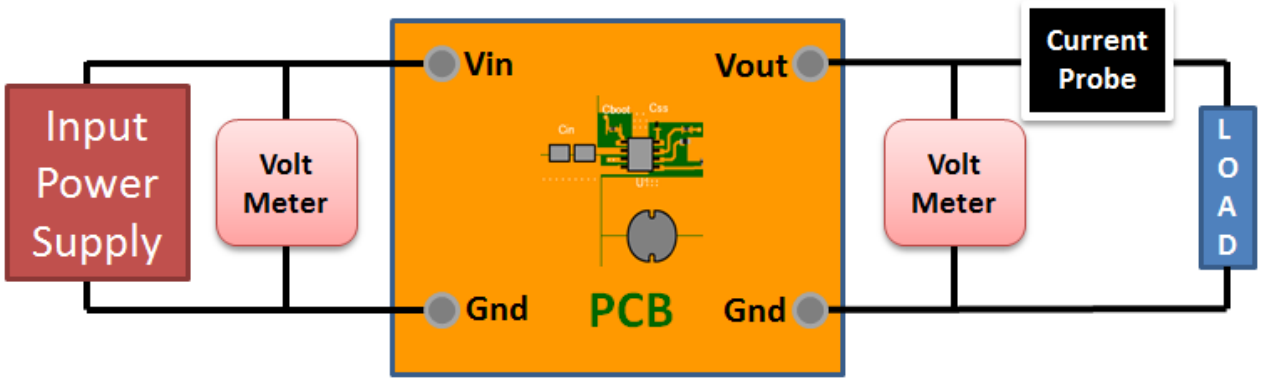
If board assembly is done in house it is best to tack down one terminal of a component on the board then solder the other terminal. For surface mount parts with large tabs, such as the DPAK, the tab on the back of the package should be pre-tinned with solder, then tacked into place by one of the pins. To solder the tab town to the board place the iron down on the board while resting against the tab, heating both surfaces simultaneously. Apply light pressure to the top of the plastic case until the solder flows around the part and the part is flush with the PCB. If the solder is not flowing around the board you may need a higher wattage iron (generally 25W to 30W is enough).

Initial Startup of Circuit

It is best to initially power up the board by setting the input supply voltage to the lowest operating input voltage 9.0V and set the input supply's current limit to zero. With the input supply off connect up the input supply to Vin and GND. Connect a digital volt meter and a load if needed to set the minimum Iout of the design from Vout and GND. Turn on the input supply and slowly turn up the current limit on the input supply. If the voltage starts to rise on the input supply continue increasing the input supply current limit while watching the output voltage. If the current increases on the input supply, but the voltage remains near zero, then there may be a short or a component misplaced on the board. Power down the board and visually inspect for solder bridges and recheck the diode and capacitor polarities. Once the power supply circuit is operational then more extensive testing may include full load testing, transient load and line tests to compare with simulation results.

Load Testing

The setup is the same as the initial startup, except that an additional digital voltmeter is connected between Vin and GND, a load is connected between Vout and GND and a current meter is connected in series between Vout and the load. The load must be able to handle at least rated output power + 50% (7.5 watts for this design). Ideally the load is supplied in the form of a variable load test unit. It can also be done in the form of suitably large power resistors. When using an oscilloscope to measure waveforms on the prototype board, the ground leads of the oscilloscope probes should be as short as possible and the area of the loop formed by the ground lead should be kept to a minimum. This will help reduce ground lead inductance and eliminate EMI noise that is not actually present in the circuit.



Design Assistance

1. Master key : C0723DD09D79F8AFBAB58B2D8AAED5F5[v1]
2. **LM51571** Product Folder : http://www.ti.com/product/LM51571 : contains the data sheet and other resources.

Copyright © 2026, Texas Instruments Incorporated

Important Notice and Disclaimer

TI provides technical and reliability data (including datasheets), design resources (including reference designs), application or other design advice, web tools, safety information, and other resources AS IS and with all faults, and disclaims all warranties. These resources are intended for skilled developers designing with TI products. You are solely responsible for (1) selecting the appropriate TI products for your application, (2) designing, validating and testing your application, and (3) ensuring your application meets applicable standards, and any other safety, security, or other requirements.

These resources are subject to change without notice. TI grants you permission to use these resources only for development of an application that uses the TI products described in the resource. Other reproduction and display of these resources is prohibited. No license is granted to any other TI intellectual property right or to any third party intellectual property right. TI disclaims responsibility for, and you will fully indemnify TI and its representatives against, any claims, damages, costs, losses, and liabilities arising out of your use of these resources.

Providing these resources does not expand or otherwise alter TI's applicable Terms of Sale or other applicable terms available either on ti.com or provided in conjunction with TI products.